Machine Learning I

Bjoern Andres, Lucas Fabian Naumann, Emil Powierski

Machine Learning for Computer Vision
TU Dresden



https://mlcv.inf.tu-dresden.de/courses/26-winter/ml1/

Winter Term 2026/2027

- A **linear program (LP)** is an optimization problem where the goal is to minimize (or maximize) a linear objective function, subject to a set of linear equality and inequality constraints.
- Below is given an example of an LP:

$$\begin{aligned} \min_{x_1,x_2,x_3,x_4 \in \mathbb{R}} \quad & x_1 - 3x_2 + 2x_3 \\ \text{subject to} \quad & x_1 + x_2 \geq 2 \\ & 3x_2 - x_3 + x_4 = 5 \\ & x_3 + x_4 \leq 3 \\ & x_1 \geq 0 \\ & x_2 \leq 0 \end{aligned}$$

- x_1, x_2, x_3, x_4 are called **decision variables**.
- The **objective function** is linear and it can be written as $c^T x$, where $c^T = (1, -3, 2, 0)$ is called **cost vector**.
- The **constraints** are linear equalities and inequalities, and can be written in the form $a^T x = b$, $a^T x \leq b$, or $a^T x \geq b$.
- The second constraint is of the form $a^T x = b$, with $a^T = (0, 3, -1, 1)$, and $b = 5$.

– In general, an **LP** is given by:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^T x \\ \text{subject to} \quad & a_i^T x \geq b_i \quad i \in M_1 \\ & a_i^T x \leq b_i \quad i \in M_2 \\ & a_i^T x = b_i \quad i \in M_3 \\ & x_j \geq 0 \quad j \in J_1 \\ & x_j \leq 0 \quad j \in J_2 \end{aligned}$$

– **Decision variables**: $x_1, x_2, \ldots, x_n$,

– **Cost vector**: $c \in \mathbb{R}^n$,

– **Objective function**: $c^T x$,

– **Linear constraints**: defined by $a_i \in \mathbb{R}^n$ and $b_i \in \mathbb{R}$, for $i \in M_1 \cup M_2 \cup M_3$,

– **Non-negativity, non-positivity constraints**: $x_j \geq 0$ for $j \in J_1$ and $x_j \leq 0$ for $j \in J_2$.

– In general, an **LP** is given by:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^T x \\ \text{subject to} \quad & a_i^T x \geq b_i \quad i \in M_1 \\ & a_i^T x \leq b_i \quad i \in M_2 \\ & a_i^T x = b_i \quad i \in M_3 \\ & x_j \geq 0 \quad j \in J_1 \\ & x_j \leq 0 \quad j \in J_2 \end{aligned}$$

– A **feasible solution** is a vector x satisfying all constraints.

– The **cost** of a feasible solution x is $c^T x$.

– An **optimal solution** is a feasible solution x^* that minimizes the objective function. Formally, $c^T x^* \leq c^T x$, $\forall x$ feasible.

– The **optimal cost** is the value $c^T x^*$.

– A decision variable x_j is called **unconstrained** if $j \notin J_1 \cup J_2$.

A farm purchases two types of feed.
It wants to meet given nutrient requirements at minimum total cost.

The nutrient content per kilogram and the minimum requirements are:

	Feed 1	Feed 2	Minimum required
Protein	200	100	400
Fibre	100	300	600
Cost (EUR/kg)	2	3	

Task: Construct the corresponding LP!

Decision variables:

$$x_1 = \text{kilograms of Feed 1 purchased},$$
$$x_2 = \text{kilograms of Feed 2 purchased}.$$

Objective: Minimize the total cost:

$$2x_1 + 3x_2.$$

Constraints: Meet the minimum nutrient requirements and purchase nonnegative quantities.

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & 2x_1 + 3x_2 \\ \text{subject to} \quad & 200x_1 + 100x_2 \geq 400 \qquad \text{(protein)} \\ & 100x_1 + 300x_2 \geq 600 \qquad \text{(fibre)} \\ & x_1, x_2 \geq 0. \end{aligned}$$

- By multiplying the objective function by -1, we can **convert a maximization problem into a minimization problem** and vice versa. Thus, we can assume that the objective function is always to be minimized.
- By multiplying inequality constraints by -1, we can **convert $\leq$ constraints into $\geq$ constraints** and vice versa. Thus, we can assume that all inequality constraints are $\geq$ constraints:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^T x \\ \text{subject to} \quad & \\ & a_i^T x \geq b_i \quad i \in M_1 \\ & a_i^T x = b_i \quad i \in M_2 \\ & x_j \geq 0 \quad\quad j \in J \end{aligned}$$

A company produces two juice blends by mixing apple and orange juice. It wants to **maximize** its total revenue.

The requirements per litre and the available supplies are:

	Blend 1	Blend 2	Available
Apple juice (litres)	0.75	0.25	300
Orange juice (litres)	0.25	0.75	240
Revenue (EUR/litre)	3	4	

Task: Construct the corresponding LP!

Decision variables:

$x_1 =$ litres of Blend 1 produced, $\quad x_2 =$ litres of Blend 2 produced.

Objective: Maximize the total revenue

$$3x_1 + 4x_2.$$

Constraints: Production cannot exceed the available supplies, and quantities must be nonnegative

$$\begin{aligned} \max_{x \in \mathbb{R}^2} \quad & 3x_1 + 4x_2 \\ \text{subject to} \quad & 0.75x_1 + 0.25x_2 \leq 300 \qquad \text{(apple juice)} \\ & 0.25x_1 + 0.75x_2 \leq 240 \qquad \text{(orange juice)} \\ & x_1, x_2 \geq 0. \end{aligned}$$

Equivalently:

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & -3x_1 - 4x_2 \\ \text{subject to} \quad & -0.75x_1 - 0.25x_2 \geq -300 \qquad \text{(apple juice)} \\ & -0.25x_1 - 0.75x_2 \geq -240 \qquad \text{(orange juice)} \\ & x_1, x_2 \geq 0. \end{aligned}$$

Definition 1. For any $m, n \in \mathbb{N}_0$, any $A \in \mathbb{R}^{m \times n}$, any $A' \in \mathbb{R}^{m' \times n}$, any $b \in \mathbb{R}^m$, any $b' \in \mathbb{R}^{m'}$ and any $J \subseteq n$, the **general system** defined by A, A', b, b' and J is

$$A\,x = b \tag{1}$$

$$A'\,x \geq b' \tag{2}$$

$$\forall j \in J\colon \quad x_j \geq 0 \ . \tag{3}$$

Definition 2. For any general system defined by A, A', b, b' and J, and any $c \in \mathbb{R}^n$, the **linear optimization problem** or **linear program (LP)** in **general form** is

$$\min \{c^T x \mid A\,x = b \ \wedge \ A'\,x \geq b' \ \wedge \ (\forall j \in J\colon x_j \geq 0) \ \wedge \ x \in \mathbb{R}^n\} \ . \tag{4}$$

Definition 3. For any $A \in \mathbb{R}^{m \times n}$ and any $b \in \mathbb{R}^m$, the **canonical system** defined by A and b is

$$A\,x \geq b \tag{5}$$
$$x \geq 0 \ . \tag{6}$$

Definition 4. For any $A \in \mathbb{R}^{m \times n}$ and any $b \in \mathbb{R}^m$, the **standard system** defined by A and b is

$$A\,x = b \tag{7}$$
$$x \geq 0 \ . \tag{8}$$

Lemma 1. General, canonical, and standard systems are mutually equivalent, in the sense that each can be transformed into any of the others.

Proof. Every canonical system is a general system.

To turn any general system into an equivalent canonical system, proceed as follows:

- For any $j \in n \setminus J$, replace the unconstrained variable x_j by $x_j^+ - x_j^-$ such that

$$x_j^+ \geq 0 \tag{9}$$

$$x_j^- \geq 0 \ . \tag{10}$$

- Replace any equality $A_{i\cdot}\, x = b_i$ by the inequalities

$$A'_{i\cdot}\, x \geq b_i \tag{11}$$

$$-A'_{i\cdot}\, x \geq -b_i \ . \tag{12}$$

Every standard system is a general system.

To turn any general system into an equivalent standard system, proceed as follows:

- Replace all unconstrained variables as described above.
- Replace any inequality $A'_{i\cdot}\, x \geq b_i$ by

$$A_{i\cdot}\, x - s_i = b_i \tag{13}$$

$$s_i \geq 0 \ . \tag{14}$$

□

Task. (a) Classify the given system as general, canonical or standard. (b) Give an equivalent canonical system. (c) Give an equivalent standard system.

System:

$$\begin{aligned} x + y + z &= 10, \\ 2x + z &= 0, \\ x, y, z &\geq 0. \end{aligned}$$

(a) Standard form: all constraints are equalities or nonnegativity constraints.

It is also a special case of a general system.

(c) The given system already has the required standard form. No transformation is needed.

(b) Canonical form

$$\begin{aligned} x + y + z &\geq 10, \\ -x - y - z &\geq -10, \\ 2x + z &\geq 0, \\ -2x - z &\geq 0, \\ x, y, z &\geq 0. \end{aligned}$$

Replace each equality by two inequalities:

$$a^\top u = b \quad \Longleftrightarrow \quad \begin{cases} a^\top u \geq b, \\ -a^\top u \geq -b. \end{cases}$$

All variables are already nonnegative.

Task. (a) Classify the given system as general, canonical or standard. (b) Give an equivalent canonical system. (c) Give an equivalent standard system.

System:

$$\begin{aligned} x + y &\geq 5, \\ -x + 2y &\geq -3, \\ 3x + 2y &= 12, \\ y &\geq 0. \end{aligned}$$

(a) General form: mixed equalities and inequalities, with x unrestricted.

For (b) and (c), write

$$x = x^+ - x^-, \qquad x^+, x^- \geq 0.$$

For (c), introduce surplus variables $s_1, s_2 \geq 0$.

(b) Canonical form

$$\begin{aligned} x^+ - x^- + y &\geq 5, \\ -x^+ + x^- + 2y &\geq -3, \\ 3x^+ - 3x^- + 2y &\geq 12, \\ -3x^+ + 3x^- - 2y &\geq -12, \\ x^+, x^-, y &\geq 0. \end{aligned}$$

(c) Standard form

$$\begin{aligned} x^+ - x^- + y - s_1 &= 5, \\ -x^+ + x^- + 2y - s_2 &= -3, \\ 3x^+ - 3x^- + 2y &= 12, \\ x^+, x^-, y, s_1, s_2 &\geq 0. \end{aligned}$$